

UAV Digital Interface System

Technical Development & Simulation Report — Phase 1

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1. EXECUTIVE SUMMARY

This report documents a complete UAV simulation and autonomous control system built on PX4 Autopilot, ROS 2 Jazzy, Gazebo Harmonic, MAVSDK-Python, and QGroundControl on Ubuntu 24.04 LTS. The project progressed from initial environment setup through autonomous multi-waypoint missions, real-time telemetry, camera integration, and optical flow-based obstacle avoidance research.

What Was Built

A fully operational UAV digital interface system capable of autonomous waypoint navigation, real-time telemetry streaming from PX4 to ROS 2 at 100 Hz, camera-based obstacle detection, and aerial photography. The system runs entirely in simulation but demonstrates integration patterns applicable to real hardware. Key components include MAVSDK-Python for mission control, Micro XRCE-DDS Agent (built from source) for PX4-ROS2 bridging, and OpenCV optical flow processing for reactive navigation.

Key Achievements

- PX4 Autopilot and Gazebo Harmonic installed and integrated on Ubuntu 24.04
- Micro XRCE-DDS Agent built from source with proper Fast DDS v3 bundling
- Live telemetry pipeline: PX4 → ROS 2 → Gazebo at 100 Hz
- Autonomous missions with GPS auto-sync (works across any Gazebo world)
- Camera integration with real-time video feed to Gazebo Image Display
- Aerial photography capture via window automation (xdotool + ImageMagick)
- Optical flow-based obstacle detection at ~25 FPS
- Multi-waypoint mission execution at 150m, 300m, and 500m
- Reactive obstacle avoidance using PositionNedYaw control
- Over 40 integration issues identified and resolved

Completion Status: 16 milestones at 100%, 4 actively in progress (obstacle avoidance at 65%, custom UI at 40%).

2. SYSTEM ARCHITECTURE

Hardware & OS

The development system is an HP ENVY x360 2-in-1 laptop with Intel Core i7 (13th Gen), 16GB LPDDR5 RAM, integrated Intel Iris Xe graphics, and 512GB NVMe SSD. Ubuntu 24.04 LTS runs on Wayland with X11 fallback for Gazebo compatibility. Integrated graphics required HEADLESS=1 mode in some scenarios to prevent GUI lag.

Software Stack

The system integrates multiple open-source components. PX4 v1.15.x handles flight control and autonomous navigation. ROS 2 Jazzy Jalisco serves as middleware for sensor/actuator communication across the stack. Gazebo Harmonic LTS provides 3D physics simulation and environment visualization. MAVSDK-Python offers a high-level Python API for mission scripting. Micro XRCE-DDS Agent v2.4.3, built from source with bundled Fast DDS v3, bridges PX4 uORB topics to ROS 2. QGroundControl provides mission planning and real-time telemetry monitoring. OpenCV and FFmpeg handle image processing and video streaming.

Communication

MAVLink runs over UDP—port 14540 for PX4 ↔ MAVSDK, port 18570 for PX4 ↔ QGroundControl. The Micro XRCE-DDS Agent bridges PX4 via UDP 8888 to ROS 2 using the uXRCE-DDS protocol. Gazebo publishes camera images through internal transport to the /camera topic. This architecture decouples mission control, telemetry, and sensor processing into separate processes that communicate via well-defined protocols.

3. DEVELOPMENT MILESTONES

Installation & Basic Setup

PX4 installation was straightforward using the official setup script with `--no-nuttx` flag. The real effort came from Ubuntu 24.04-specific compatibility issues. Gazebo Harmonic required installation from the OSRF repository (not default Ubuntu packages, which included obsolete Classic). X11 fallback was essential because Wayland caused intermittent rendering glitches on integrated graphics. ApplImage compatibility required `libfuse2t64`, which wasn't on the system by default. ROS 2 Jazzy installation followed the official playbook with locale setup, GPG key registration, and repository pinning.

The Micro XRCE-DDS Saga

This was the biggest integration challenge. The agent isn't packaged for Ubuntu 24.04 yet, so it had to be built from source. Initial attempts with `colcon` build failed—Fast DDS v3 wasn't installed, and the `-DTHIRDPARTY=ON` flag didn't get passed correctly through the `colcon` wrapper. Spent hours debugging CMake output only to realize `colcon` wasn't propagating arguments to the underlying build system. Switching to direct CMake with the `-DTHIRDPARTY=ON` flag worked immediately. Lesson: when a build system abstraction layer doesn't cooperate, bypass it.

Once built, the agent needed careful orchestration. Starting it after PX4 caused "Running, disconnected" messages. Restarting it mid-simulation crashed PX4 with timeout errors. The solution was strict startup order: Agent first, PX4 second, ROS 2 nodes third. Kill stale processes before each run: `pkill -9 px4`; `pkill -9 ruby`; `pkill -9 gz`; `sudo fuser -k 8888/udp`.

QoS Matching

After building `px4_msgs` from source, ROS 2 topics should have been immediately available. They weren't. `ros2 topic echo` returned nothing. Spent an hour checking subscriptions, XML files, and middleware configuration. The culprit: QoS mismatch. PX4 publishes high-frequency topics (odometry, IMU) with Best Effort reliability—because resending stale sensor data is worse than waiting for fresh readings. `ros2 topic echo` defaults to Reliable QoS, which doesn't match. The fix was one flag: `--qos-reliability best_effort`. After that, 100 Hz telemetry streamed without drops. This was a critical lesson about ROS 2 QoS policies and how they silently fail when incompatible.

MAVSDK Mission Development

Building autonomous missions revealed four distinct problem-solution cycles. Attempt 1 used positional arguments for `MissionItem`, which failed with a `TypeError` because the parameter signature changed between MAVSDK versions. Switched to keyword arguments—forward compatible. Attempt 2 succeeded in uploading but the UAV skipped directly to RTL, never visiting waypoints. Added `drone.action.takeoff()` before `mission.start_mission()`. Attempt 3 worked but waypoints were 9,384 km away because Gazebo was set to Baylands (California) while PX4 defaulted to Zurich. Hardcoding GPS

coordinates fails across worlds. The solution: read GPS from live telemetry and compute waypoints relative to it. Missions are now world-agnostic—port them to any Gazebo world without modification.

Camera Integration & The Capture Saga

Adding a camera to the x500 model required SDF modification. A simple camera link pointing forward at 0.12m ahead, publishing to /camera at 30 Hz. Expected the GStreamer plugin to work. It doesn't exist in Ubuntu 24.04 Harmonic. Switched to native Image Display—works fine for SITL. Aerial photography was the real battle. Tried 20+ approaches: Gazebo services wrote 0-byte files. OpenCV UDP streams had port conflicts. FFmpeg pipelines timed out. GStreamer was missing plugins. gnome-screenshot captured the terminal instead. ImageMagick alone couldn't find the right window. The working solution was ugly but reliable: xdotool finds the Gazebo window, ImageMagick captures it, subprocess pipes everything together. It works. It's not elegant. But it captures photos on demand during missions.

Coordinate Synchronization (The 9M Problem)

QGroundControl initially showed Zurich (Milchbuck) while Gazebo rendered Baylands, California. Waypoints were positioned 9,384,145 meters away. First fix: set PX4_HOME_LAT, PX4_HOME_LON, PX4_HOME_ALT before launch. Better fix: implement bidirectional GPS-to-local and local-to-GPS conversions. This revealed a critical math bug: double $\cos(\text{latitude})$ division when converting longitude meters to degrees. The correct formula is $\text{east} = \text{dlon_degrees} * 111319.5 * \cos(\text{lat})$. The $\cos(\text{lat})$ factor must be applied exactly once. It was being applied twice, creating massive errors. After fixing this and implementing proper coordinate transforms, missions work correctly regardless of Gazebo world.

4. OBSTACLE AVOIDANCE SYSTEM

Obstacle avoidance evolved through five distinct design iterations. Each attempt uncovered different failure modes, control-system requirements, and sensor limitations. The current approach (Attempt 5) uses optical flow for reactive navigation without prior obstacle maps.

The Detection Pipeline

The system uses Lucas-Kanade sparse optical flow with Shi-Tomasi corner detection. Features (image corners) are detected, tracked across frames, and analyzed for divergence patterns—expansion indicates an approaching obstacle. The frame is divided into left, center, and right regions; divergence scores are computed separately for each region. Multi-frame confirmation (3 of 5 frames) reduces false positives during takeoff when the landing gear creates spurious motion. Parameters: maxCorners=200, qualityLevel=0.3, minDistance=7 for feature detection. LK tracking uses a 15×15 window with 2 pyramid levels. Flow magnitudes below 0.5 pixels are ignored. A median filter removes outliers. If fewer than 6 features survive validation, no obstacle is reported—optical flow is unreliable with insufficient features.

Five Iterations of Failure & Learning

Attempt 1 (Naive Body-Frame Velocity) issued direct body-frame velocity commands with yaw rate control. The UAV descended immediately because NED convention inverts the Z-axis (negative is up). Control failed. Attempt 2 added `is_airborne` checks to suppress detection during takeoff but positioned waypoints too close to origin—the UAV never reached obstacles. Attempt 3 fixed GPS-to-local conversions but flew directly through obstacles because optical flow detection came too late. Attempt 4 combined known obstacle maps with optical flow for proactive avoidance. This created heading oscillation loops: $168^\circ \rightarrow 149^\circ \rightarrow 102^\circ \rightarrow 35^\circ \rightarrow 336^\circ \rightarrow \text{repeat}$. Map-based logic conflicted with waypoint navigation. Scrapped it.

Attempt 5 switched to `PositionNedYaw` setpoints (absolute heading) instead of `VelocityBodyYawspeed` (yaw rate commands). This is crucial: continuous yaw rate commands cause motor saturation when heading errors exceed 30° . The PX4 attitude controller cannot track aggressive yaw rates while maintaining position. Absolute heading control is more stable. Optical flow now triggers only lateral avoidance—it doesn't override the waypoint heading. This separation of concerns prevents oscillation.

Critical Technical Discoveries

Discovery 1: NED Coordinate Frame. PX4 uses North-East-Down. Z-positive means descend, opposite of intuition. This inverted expectation caused immediate descent after takeoff. Discovery 2: Double $\cos(\text{latitude})$ Bug. Converting longitude from degrees to meters requires multiplying by $\cos(\text{latitude})$. Doing it twice produces 9M errors. Discovery 3: Optical Flow Limitations. Thin walls (0.5m) at 10m altitude don't produce enough visual features for reliable detection. Optical flow is vulnerable to featureless

surfaces and self-detection from propellers. Recommendation: maintain 5m+ clearance from known obstacles. Optical flow is a safety layer, not primary navigation.

Discovery 4: VelocityBodyYawspeed is Unstable. The MAVSDK docs and PX4 community confirm: aggressive yaw rate commands exceed motor authority. Use PositionNedYaw or VelocityNedYaw with absolute heading instead.

Current Architecture

Gazebo camera sensor → ros_gz_bridge (Image topic) → ROS 2 Node (optical flow processing) → feature detection, flow computation, regional divergence analysis → multi-frame confirmation → MAVSDK-Python Offboard PositionNedYaw setpoints → PX4 SITL. The navigation loop runs at ~25 FPS, limited by CPU. Detection latency is ~200ms. This is acceptable for slow-moving UAV avoidance maneuvers.

5. ISSUES & SOLUTIONS

Over forty distinct integration issues were encountered. The most impactful ones follow. Each revealed something about system design, Ubuntu 24.04 compatibility, ROS 2 semantics, or PX4 behavior.

Gazebo Classic Detected: Ubuntu 24.04 default PPA pulled Gazebo Classic instead of Harmonic. → *Installed from OSRF official repository. Added to sources.list.d.*

ROS 2 QoS Mismatch: ros2 topic echo returned no data from PX4 high-frequency topics. → *Used --qos-reliability best_effort flag. PX4 publishes Best Effort for sensor data.*

Micro XRCE-DDS Not in APT: ros-jazzy-micro-xrce-dds-agent package doesn't exist yet. → *Built from GitHub source with standalone CMake and -DTHIRDPARTY=ON flag.*

Fast DDS v3 Missing: colcon build failed—fastdds package not found. → *Abandoned colcon wrapper. Used direct CMake with bundled THIRDPARTY.*

Waypoints 9,384 km Away: GPS coordinates mismatched between Gazebo worlds. → *Implemented GPS auto-sync from telemetry + environment variables for home position.*

20+ Camera Capture Methods Failed: Gazebo services, OpenCV, FFmpeg, GStreamer all failed or timed out. → *Final solution: xdotool for window management + ImageMagick for capture.*

Infinite Rotation Loops: UAV spun uncontrollably during avoidance (heading: 168° → 35° → 336° → repeat). → *Removed map-based avoidance, switched to PositionNedYaw, limited yaw rates.*

Immediate Descent After Takeoff: MAVSDK vertical velocity sign inverted in NED frame. → *Reversed altitude control sign. Stabilized autonomous flight.*

Key Learnings

- Ubuntu 24.04 enforces venv isolation. System-wide pip install is blocked. Plan for this.
- MAVSDK parameter counts change between versions. Always use keyword arguments, never positional.
- Hardcoded GPS coordinates fail across different Gazebo worlds. GPS auto-sync is essential.
- ROS 2 topic names with leading slash publish to global namespace. Affects bridging.
- Plugin library names changed in Ubuntu 24.04. Docs may not match the system.
- PX4 requires explicit takeoff before waypoint missions. Mission sequencing matters.
- Gazebo Harmonic uses "gz" command, not "gazebo".
- HEADLESS=1 essential for integrated graphics to prevent GUI lag.
- uXRCE-DDS Agent build: standalone CMake more reliable than colcon wrapper.
- PX4 Best Effort QoS for high-frequency topics. ROS 2 subscribers must match.

- Startup order critical: Agent → PX4 → ROS 2 nodes. Wrong order = handshake failures.
- VelocityBodyYawspeed with yaw rate causes motor saturation. Use position setpoints.
- Only apply $\cos(\text{latitude})$ once when converting longitude. Double application = 9M errors.
- Maintain 5m+ clearance from obstacles. Reactive avoidance is backup, not primary.

6. PERFORMANCE RESULTS

System benchmarks show solid performance on integrated graphics. Build times are fast. Mission execution is reliable.

PX4 SITL boots in ~25 seconds. Gazebo launches in ~15 seconds. Topic discovery completes in ~5 seconds. Telemetry streams at a consistent 100 Hz from PX4 to ROS 2. Simulation real-time factor is ~0.98 (nearly real-time). Mission upload takes ~2 seconds. Optical flow processing runs at ~25 FPS—CPU bound due to integrated graphics. Obstacle detection latency is ~200ms from camera frame to avoidance command.

Multi-waypoint missions executed perfectly. The UAV reached each waypoint within meters, maintained stable altitude, captured aerial photos at the final waypoint, and returned to launch without issues. Telemetry logs showed clean sensor data, proper mode transitions, and expected control behavior throughout.

Milestone Completion: 16 milestones at 100% (installation, telemetry, missions, camera, coordinate sync, photo capture, etc.). 4 milestones in progress (obstacle avoidance 65%, custom UI 40%).

7. CODE & ARCHITECTURE

Three major code artifacts drive the system: the autonomous mission script, the optical flow avoidance node, and the modified x500 model SDF.

Mission Script (`mission.py`)

MAVSDK-based Python script that uploads waypoints, arms the UAV, executes the mission, and captures aerial photos. Uses `async/await` for non-blocking telemetry. GPS auto-sync reads current position from telemetry and computes waypoints relative to it. Window automation (`xdotool` + `ImageMagick`) handles photo capture at the final waypoint. Graceful error handling for missing files. Supports headless operation.

Obstacle Avoidance Node (`optical_flow_avoidance.py`)

ROS 2 node that subscribes to camera images, processes optical flow in real-time, and generates avoidance commands. Features auto-retry camera topic subscription (the bridge may start after the node). Lucas-Kanade tracking with Shi-Tomasi corner detection. Regional divergence analysis (left/center/right). Temporal smoothing with multi-frame confirmation. Outputs `PositionNedYaw` setpoints for stable heading control.

Modified SDF Model

The x500 model was extended with a camera link and fixed joint. Camera points forward at 0.12m ahead, publishes to `/camera` at 30 Hz with 640×480 resolution and 60° horizontal FOV. SDF includes rotor motor plugins and motor failure simulation plugin for future testing.

8. CURRENT STATUS & NEXT STEPS

What Works

- Full PX4 + Gazebo Harmonic + QGroundControl stack
- Micro XRCE-DDS Agent from source with live telemetry
- ROS 2 receiving PX4 data at 100 Hz with correct QoS
- Autonomous multi-waypoint missions with GPS auto-sync
- Camera integration and real-time video feed
- Mission monitoring via QGroundControl
- Coordinate synchronization for any Gazebo world
- Aerial photography at waypoints
- Optical flow detection at 25 FPS
- Bidirectional GPS ↔ local coordinate conversion

In Progress

Obstacle avoidance is at 65% completion. Optical flow detection works reliably. Regional analysis identifies obstacle direction. Current work: validating PositionNedYaw control strategy in SITL and ensuring clean recovery after obstacles. Custom ROS 2 interface node (40% complete) will provide RViz2 visualization and dashboard monitoring.

Roadmap

- **Phase 1.1 — Custom ROS 2 interface node + dashboard:** May 2026 / RViz2 integration
- **Phase 1.2 — Advanced flight modes (precision landing):** May 2026 / Python scripts
- **Phase 1.3 — Validated obstacle avoidance in SITL:** May 2026 / Demo video
- **Phase 1.4 — GStreamer video bridge to QGC:** June 2026 / UDP streaming
- **Phase 3 — Hardware-in-the-loop testing:** July 2026 / Hardware integration
- **Phase 4 — Portfolio & documentation:** July 2026 / LinkedIn + GitHub

9. APPENDICES

Essential Commands

Daily workflow:

Source ROS 2 and drone environment. Kill stale processes. Export home coordinates. Launch PX4 SITL. Run missions. The startup order is critical: Micro XRCE-DDS Agent first, then PX4, then ROS 2 nodes. Skipping or reversing this causes handshake failures.

Obstacle avoidance launch:

Three terminals: T1 runs `ros_gz_bridge` for camera topic bridging, T2 launches PX4 with obstacle course world, T3 executes the optical flow node. All three must be running simultaneously.

Project Structure

Source code lives in `src/` (mission scripts, ROS 2 nodes). Launch files in `launch/`. Gazebo worlds in `worlds/`. Custom models in `gazebo_models/`. Configuration in `config/`. Tests in `tests/`. Utility scripts in `scripts/`. Documentation in `docs/`.

Gazebo Worlds

Default Zurich world for testing. Baylands world for California-based missions. Custom `obstacle_course` world with 12 strategically placed obstacles for avoidance testing. Each world is defined by SDF with different coordinate origins.

Camera Capture Methods Evaluated

Gazebo services (wrote 0-byte files). OpenCV UDP (port conflicts). FFmpeg pipelines (timeouts). GStreamer (missing plugins). gnome-screenshot (captured wrong window). ImageMagick alone (couldn't find window). Final winner: `xdotool` + ImageMagick. It's reliable and production-ready.

DOCUMENT CONTROL

Version 1.0: Initial report (core milestones). V1.1: Added PX4 and QGC setup, Ubuntu 24.04 specifics. V1.2: MAVSDK missions, camera integration, SDF mods. V1.3: Coordinate sync, camera capture saga (20+ methods). V1.4: Micro XRCE-DDS build saga, QoS debugging, live telemetry, 30+ issues. V1.5: Optical flow obstacle avoidance, control strategy evolution, coordinate frame analysis, 10 new issues, engineering discoveries.

This report documents the complete UAV Digital Interface project Phase 1 for professional portfolio use. All development verified on HP ENVY x360 running Ubuntu 24.04 LTS. All source builds (PX4, ROS 2 packages, Micro XRCE-DDS Agent) completed successfully. The system is reproducible and documented for handoff or future work.

10. DETAILED CODE WALKTHROUGH

Mission Script Architecture

The mission script is structured around `async/await` patterns for non-blocking telemetry queries. The core flow: connect → wait for GPS → read home coordinates → compute waypoints → upload → arm → execute → capture. Each step includes error handling and fallback behavior.

Key design decisions:

- Async/await: Telemetry callbacks don't block mission execution. Multiple sensors stream simultaneously.
- GPS auto-sync: Read home position from telemetry instead of hardcoding. Missions work in any Gazebo world.
- Keyword arguments: `MissionItem` parameters use named args for forward compatibility across MAVSDK versions.
- Subprocess isolation: Window automation runs in subprocess to avoid blocking the drone connection.
- Error recovery: If capture fails, `gnome-screenshot` is attempted as fallback.

Optical Flow Processing Pipeline

The optical flow node subscribes to camera images and processes them frame-by-frame. Performance is limited by CPU (integrated graphics), not by algorithm design. The pipeline has three stages: feature detection, flow computation, and regional analysis.

Stage 1: Feature Detection

Shi-Tomasi corner detection identifies strong features in the image. Parameters (`maxCorners=200`, `qualityLevel=0.3`, `minDistance=7`) are tuned for a 640×480 camera feed at 30 Hz. Lower `qualityLevel` increases sensitivity but creates more false positives. Fewer corners mean fewer trackable features and higher noise in flow estimates.

Stage 2: Lucas-Kanade Optical Flow

Sparse optical flow tracks detected features across frames. The 15×15 window is large enough to capture neighborhood context but small enough to avoid mixing foreground and background motion. Two pyramid levels handle both small and large displacements. If fewer than 6 features survive tracking, the frame is skipped—optical flow is unreliable with too few features.

Stage 3: Regional Divergence Analysis

Flow vectors are analyzed per region. The image is divided into left (0-33%), center (33-66%), and right (66-100%) sections. Median flow magnitude in each region is computed. Regional comparison indicates obstacle direction: high center score means straight-ahead obstacle; high left or right score indicates the obstacle is to that side. A threshold (2.0 pixels) filters noise. Multi-frame confirmation requires 3 of the last 5 frames to agree

before triggering avoidance. This temporal smoothing prevents jitter from a single spurious frame.

Control Integration with MAVSDK

The avoidance node computes lateral velocity commands that get wrapped into PositionNedYaw setpoints. The mission script remains in Offboard mode, continuously sending position commands. When optical flow detects an obstacle to the right, the lateral velocity command shifts the UAV right while maintaining forward progress toward the waypoint. This approach keeps the navigation loop stable—absolute heading prevents yaw rate oscillation.

Why PositionNedYaw instead of VelocityBodyYawspeed:

Velocity commands with yaw rate require the PX4 attitude controller to track a desired heading AND desired velocity simultaneously. When yaw rate is aggressive (e.g., $45^\circ/\text{s}$), heading error can exceed 30° , saturating motors. The position controller allows gradual heading changes, preventing saturation. Additionally, absolute position setpoints automatically handle model-specific dynamics (inertia, drag) that velocity setpoints require manual tuning for.

11. DEPLOYMENT & REPRODUCIBILITY

System Requirements

Minimum: Ubuntu 24.04 LTS, 8GB RAM, dual-core CPU. Recommended: 16GB RAM, quad-core i7, integrated GPU. The system was developed on an HP ENVY x360 with integrated Iris Xe graphics—not high-end hardware. Gazebo and PX4 SITL scale well from modest to powerful machines.

Setup Checklist

- 1. Install Ubuntu 24.04 LTS (clean install recommended)
- 2. Clone PX4-Autopilot repository with submodules
- 3. Run PX4 setup script with --no-nuttx flag
- 4. Add OSRF repository and install Gazebo Harmonic
- 5. Install ROS 2 Jazzy with desktop-full variant
- 6. Build Micro XRCE-DDS Agent from source (critical step)
- 7. Build px4_msgs from source in ROS 2 workspace
- 8. Install MAVSDK-Python in dedicated venv
- 9. Install QGroundControl AppImage
- 10. Clone mission and obstacle avoidance scripts
- 11. Verify all executables: px4, make, gz, ros2, python3
- 12. Test startup sequence: Agent → PX4 → ROS 2

Common Pitfalls & Fixes

PX4 build fails on source fetch: Clone with --recursive flag: `git clone https://github.com/PX4/PX4-Autopilot.git --recursive`

Gazebo Harmonic not found (Classic installed instead): Add OSRF PPA: `sudo apt-add-repository ppa:osrf/simulation && sudo apt update`

ROS 2 topics invisible (no data): Check QoS: `ros2 topic echo --qos-reliability best_effort /fmu/out/vehicle_odometry`

MicroXRCEAgent "Running, disconnected": Start Agent BEFORE PX4. Kill stale processes: `pkill -9 px4; pkill -9 gz; sudo fuser -k 8888/udp`

MAVSDK connection "Name does not resolve": Use `udpin://0.0.0.0:14540` instead of `localhost`. `0.0.0.0` binds to all interfaces.

Camera topic /camera not found: Verify SDF model includes camera sensor. Check: `gz topic -l | grep camera`

Waypoints 9000+ km away: Set `PX4_HOME_LAT`, `PX4_HOME_LON`, `PX4_HOME_ALT` before launch

Gazebo window doesn't render (Wayland): Log in to X11 session or set `HEADLESS=1` for CLI-only operation

12. ENGINEERING INSIGHTS & TECHNICAL DEEP DIVES

ROS 2 QoS: The Silent Failure

QoS (Quality of Service) policies determine how ROS 2 nodes negotiate subscription compatibility. PX4 publishes odometry at 100 Hz with Best Effort reliability—this means: "send once per cycle; if the subscriber misses it, that's OK." This is correct for high-frequency sensor data because stale information is worse than no information. A subscriber expecting Reliable QoS (guaranteed delivery) cannot receive from a Best Effort publisher. There is no error message. The subscription silently fails. Spent hours debugging XML, checking middleware configuration, and inspecting topic definitions before discovering this mismatch.

The lesson: ROS 2 QoS mismatches fail silently. Always verify with `ros2 topic info /topic_name` to see the QoS profile, then explicitly match it in subscribers.

Coordinate Frame Conventions: A Source of Subtle Bugs

The system uses four coordinate frames: ENU (Gazebo world), NED (PX4), Body (UAV-relative), and GPS. Conversions between frames are error-prone. ENU uses East-North-Up (X=East, Y=North, Z=Up). NED uses North-East-Down (X=North, Y=East, Z=Down). Body frame uses Forward-Right-Down. Each frame has different sign conventions and axis orderings.

A common mistake: treating NED Z-axis like an altitude. In NED, positive Z is down (descending). A MAVSDK altitude command of +2.0 means "move down 2 meters," not up. This inverted intuition caused the UAV to dive immediately after takeoff. Another mistake: applying $\cos(\text{latitude})$ twice when converting longitude. The math is: $\text{east_meters} = \text{dlon_degrees} * 111319.5 * \cos(\text{lat})$. The $\cos(\text{lat})$ factor accounts for longitude degree width decreasing near the poles. It should be applied exactly once. Double application creates 9M errors.

Best practice: Always explicitly document which frame you're working in. Use clear variable names: `north_meters`, `east_meters`, `down_meters` vs `lat_deg`, `lon_deg`, `alt_m`. Test frame conversions independently before integrating them into the navigation loop.

Why Optical Flow Fails on Featureless Surfaces

Optical flow requires detectable features to track. A blank white wall has no corners or edges—nothing to track. The Shi-Tomasi corner detector fails. A thin wall (0.5m) at 10m altitude may not fill enough pixels to produce reliable features. A propeller spinning in the camera FOV creates spurious features that move in ways the algorithm doesn't expect, producing false positives. Textured obstacles (trees, buildings, rocks) work well. Featureless obstacles fail.

This is why optical flow alone is insufficient for safe navigation. It requires either textured environments or fusion with other sensors (lidar, stereo vision, terrain maps). In this

project, optical flow is a safety layer—reactive avoidance only. The primary navigation relies on GPS waypoints, which don't require visual features.

Motor Saturation & Control Authority

PX4 motor controllers have maximum angular velocity limits ($\sim 1000^\circ/\text{s}$ for the x500 airframe). When the attitude controller tries to achieve a large heading error while maintaining altitude, it may request yaw rates that exceed motor limits. The motors saturate. The aircraft loses control authority and can't track the commanded heading or velocity. This manifests as spinning loops.

VelocityBodyYawspeed commands with aggressive yaw rates ($>30^\circ/\text{s}$) trigger this saturation easily. PositionNedYaw with absolute heading allows gradual heading changes, staying within motor authority. The position controller automatically manages rate limits internally.

Lesson: Use position setpoints (PositionNedYaw) for stable control. Reserve velocity setpoints (VelocityBodyYawspeed) for low-level experimentation only. The extra layer of abstraction is worth it.

13. INTEGRATION PATTERNS & ARCHITECTURAL DECISIONS

Multi-Process Architecture

The system runs four separate processes: (1) PX4 SITL—flight control, (2) Gazebo—physics simulation, (3) ROS 2 nodes—sensor processing, (4) MAVSDK mission script—high-level control. Each process is independent. They communicate via well-defined protocols (MAVLink, DDS, Gazebo Transport). This architecture provides several benefits:

- Fault isolation: If one process crashes, others continue running
- Resource separation: CPU-heavy Gazebo doesn't starve other processes
- Easy testing: Each component can be tested independently
- Scalability: Processes can run on separate machines over network
- Clear interfaces: Protocols force component boundaries to be explicit

The downside is orchestration complexity. The startup order matters. Communication latency is non-zero (~10-50ms per hop). Debugging requires looking at multiple log files. But these are acceptable tradeoffs for a research/portfolio project.

GPS Auto-Sync Pattern

Instead of hardcoding waypoint coordinates, the mission script reads the current position from live telemetry and computes relative waypoints. This makes missions world-agnostic. The same mission script works in Zurich, Baylands, or any custom world. The pattern:

1. Connect to drone and read current GPS position
2. Compute waypoint offsets (north, east, altitude) relative to home
3. Convert offsets back to GPS coordinates using home position
4. Upload GPS coordinates to PX4

The key insight: PX4 expects GPS coordinates, but you can derive them from relative offsets. This decouples mission planning (north 100m, east 50m) from world coordinates (latitude, longitude).

Multi-Frame Confirmation for Temporal Smoothing

Optical flow detection can produce false positives on a single frame. A spinning propeller, motion blur, or camera glitch can trigger obstacle detection when nothing is actually there. Multi-frame confirmation requires 3 of the last 5 frames to agree before triggering avoidance. This reduces false positives without significant latency increase (200ms at 30 FPS).

The pattern is general: whenever a sensor produces noisy binary decisions (obstacle present / not present), maintain a rolling window and require consensus before triggering actions. This simple temporal filtering prevents jittery behavior.

14. WHAT DIDN'T MAKE THE CUT

Several ambitious ideas were attempted but didn't ship. Worth documenting why they failed or got deprioritized.

GStreamer Camera Plugin

The goal: stream H.264-compressed video from Gazebo to QGroundControl at 10 Hz for low-bandwidth monitoring. Gazebo Harmonic should support this via the GStreamer plugin. It doesn't exist in Ubuntu 24.04 packages. Attempts to build it from source failed with missing dependencies. Workaround: use Gazebo's native Image Display plugin and xdotool capture. This works but is not the elegant solution initially envisioned.

Pre-Computed Obstacle Map

Idea: load a known obstacle map and use A* pathfinding to compute collision-free routes. Combined with optical flow for reactive avoidance, this would be proactive + reactive hybrid. Implementation: hardcoded obstacle positions in Python dict, computed safe waypoints, fed them to mission planner. Result: UAV entered uncontrollable spinning loops (Attempt 4). The map avoidance logic conflicted with waypoint-following logic. Decision: abandon the hybrid approach and stick to pure reactive optical flow (Attempt 5). Lesson: don't combine two control laws without careful verification. One always wins, and the other causes instability.

Hardware-in-the-Loop (HIL)

Goal: test the SITL-developed code on a real Pixhawk drone without hardware changes. PX4 supports HIL mode where the flight controller connects to Gazebo for sensor readings instead of onboard hardware. Not attempted because: (1) No access to real Pixhawk hardware during development, (2) SITL validation is sufficient for Phase 1 portfolio piece, (3) HIL requires careful sensor timing that adds complexity. Deferred to Phase 3.

Stereo Vision Obstacle Detection

Optical flow works well for detecting motion but not absolute distance. Stereo vision (two cameras with known baseline) can compute depth without motion, enabling static obstacle detection. Not implemented because: (1) Adds computational cost (two camera feeds, disparity estimation), (2) Gazebo integration is non-trivial, (3) Single optical flow is sufficient for SITL validation. Good candidate for future work.

15. PORTFOLIO PRESENTATION GUIDE

For Technical Interviews

Interviewers want to understand (1) what you built, (2) why you made specific decisions, (3) what problems you solved, (4) what you learned. Prepare to walk through:

- System architecture: 4 independent processes, why separation of concerns matters
- Integration challenges: Micro XRCE-DDS, QoS matching, coordinate frames
- Control strategy evolution: why 4 approaches failed, what worked and why
- Technical discoveries: NED convention, double cos(latitude), optical flow limitations
- Trade-offs: SITL vs HIL, optical flow vs lidar, complexity vs robustness
- What you'd do differently: better planning upfront, more testing per iteration

For LinkedIn / Resume

Lead with impact: "Built a complete autonomous UAV control system integrating PX4, ROS 2, and Gazebo. Developed optical flow-based obstacle avoidance and autonomous mission planning with GPS auto-sync. Resolved 40+ integration issues across multi-platform stack."

Then back it up with specifics: (1) Technologies used, (2) Problem solved, (3) Technical metrics (100 Hz telemetry, 25 FPS optical flow, 150-500m autonomous missions). Include a GitHub link if repository is public. Link to this technical report.

For Demos

If demoing live: (1) Pre-warm Gazebo and PX4 (launch takes 40 seconds). (2) Run a simple 150m waypoint mission—shortest mission, least things that can go wrong. (3) Show QGroundControl telemetry overlay. (4) Show mission planning in QGC. (5) Show camera feed. (6) If time: show obstacle avoidance demo (more complex, more points of failure).

If demoing with video: Record a 2-3 minute montage: (1) System startup (fast-forwarded), (2) Mission planning in QGC, (3) Autonomous flight, (4) Telemetry streaming, (5) Final photo capture, (6) Obstacle avoidance in custom world. Narrate the "why" not just the "what."

16. TESTING STRATEGY & VALIDATION

Unit Testing Approach

Python scripts don't require compilation, so testing is quick. Created simple test scripts for:

- Mission upload: verify waypoints are correctly formatted before sending to PX4
- GPS conversion: test bidirectional GPS ↔ local coordinate transformation with known values
- Optical flow thresholding: feed synthetic flow vectors and verify region detection
- Command execution: dry-run mission script without connecting to drone

No formal unit test framework (pytest) was set up—project scope didn't require it. But the validation approach was systematic: test each component independently before integration.

Integration Testing

Integration tests run the full stack: Agent → PX4 → ROS 2 → Mission. Validated:

- Telemetry streaming: confirm 100 Hz odometry, battery status, vehicle status
- Mission execution: 150m, 300m, 500m waypoints reach within 5m of target
- Coordinate sync: same mission works in Zurich and Baylands worlds
- Camera streaming: /camera topic publishes 30 FPS without drops
- Photo capture: xdotool successfully captures image at final waypoint
- Obstacle avoidance: UAV detects and avoids placed obstacles

Regression Testing

After each major change, ran the same mission twice to verify consistency. Minor code changes (parameter tuning) were tested once. Major changes (control law, coordinate frame) were tested thrice. This informal approach was sufficient for a research project but wouldn't scale to production code.

17. FINAL REFLECTIONS & LESSONS LEARNED

This project was a deep dive into robotics stack integration. Several observations stand out:

Planning vs. Exploration

Spending time upfront to understand coordinate frames, QoS semantics, and control authority would have prevented hours of debugging. But discovering these lessons through failure builds intuition that planning alone can't provide. The project benefited from a hybrid: high-level plan (what to build), low-level exploration (how to build it).

Integration is Harder Than Components

Writing a simple MAVSDK mission script is straightforward. Making it work end-to-end with PX4, ROS 2, Gazebo, and QGroundControl revealed 40+ non-obvious issues. Most were integration boundaries: API incompatibilities, version mismatches, QoS policies, protocol details. Component-level testing is valuable, but integration testing is where real learning happens.

Documentation & Code Clarity Matter

Early code was tightly coupled and hard to debug. Later iterations separated concerns: mission planning, telemetry, optical flow processing. Clear variable names (home_lat vs lat), explicit coordinate frames (ENU vs NED), and documented assumptions made troubleshooting dramatically faster. This report itself forced clarity—explaining the system to an external audience revealed gaps in understanding.

Failing Fast > Delayed Perfection

Multiple control strategies (Attempts 1-5 for obstacle avoidance) were iterated quickly. Rather than spending weeks planning the perfect control law, building fast prototypes revealed failure modes that research papers don't mention. Iteration > analysis (for this scope).

Open Source Ecosystem is Powerful

PX4, ROS 2, Gazebo, MAVSDK are all mature open-source projects. Issues were usually user error or integration gaps, not bugs. Community forums and GitHub issues provided answers. This project would be 10x harder without this ecosystem.

Hardware Constraints Shape Software Design

Integrated graphics (no dedicated GPU) limited Gazebo rendering and optical flow FPS. Instead of fighting the constraint, the design adapted: HEADLESS=1 mode, lower resolution optical flow, multi-frame confirmation to compensate for jitter. Good software design works within hardware reality, not against it.

18. CONCLUSION

The UAV Digital Interface system demonstrates end-to-end integration of modern robotics technologies. It combines PX4 flight control, ROS 2 middleware, Gazebo simulation, and autonomous mission planning into a functioning system. The project surfaced real integration challenges that textbooks and tutorials gloss over. Solving those challenges required understanding not just individual components but how they interact at boundaries.

Phase 1 is feature-complete for SITL validation. The autonomous mission pipeline works reliably. Telemetry streams at expected rates. Optical flow detection functions, with clear limitations understood. 16 major milestones are production-ready. 4 remain in active development.

Future phases will address real hardware (Pixhawk), advanced flight modes, comprehensive obstacle avoidance, and portfolio documentation. But the foundation is solid. The system is documented, reproducible, and ready for extension.

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Report compiled May 2026. Developed on Ubuntu 24.04 LTS, HP ENVY x360. All components verified working. Source code and documentation available on request. This technical report serves as both development log and architectural reference for future work.